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Academic Year: 2025-2026 Even

Department of Artificial Intelligence and Machine Learning

Date: 13-04-2026	**Test – I**	**Max. Marks: 50**
Semester: VI	**UG**	**Duration: 120 minutes**
Course Title: BIG DATA TECHNOLOGIES		**Course Code: AI362IA**

Part – A

Q. No	Question	M	BT	CO
1.1	Give any 4 differences between traditional RDBMS and map reduce.	2	1	1
1.2	What is the purpose of Namenode in HDFS? How does it help in maintaining file metadata.?	2	1	1
1.3	A fintech company stores sensitive transaction data in Apache Hadoop using Hadoop Distributed File System. One of the Data Nodes suddenly fails due to a hardware issue, but the system continues its operation. Identify the technique used to handle data Node Failures effectively.	2	1	1
1.4	What is data locality Optimization?	2	2	1
1.5	What is awk script? Why is it used?	2	1	1

Part – B

Q. No	Question	M	BT	CO
1a	Elaborate the anatomy of file read and file write with neat diagram.	5	2	1
1b	Explain the role of two types of nodes functioning in the architecture of HDFS with a neat diagram.	5	2	2
2a	How does the Combiner function contribute to optimization in data processing?	5	2	2
2b	Why Map tasks write their output to the local disk, not to HDFS?	5	2	1
3a	How is HDFS structured, and in what situations might HDFS be less suitable?	5	1	1
3b	Summarize the characteristics of Bigdata.	5	2	1
4	A data engineering team is working with **Apache Hadoop** using the **Hadoop Distributed File System** to process large-scale log data. The system has the following disk characteristics: • Disk seek time = **10 ms** • Disk transfer rate = **100 MB/s** The team wants to optimize the performance of file storage and processing by choosing an appropriate **HDFS block size**. They aim to ensure that the overhead due to disk seek time is minimal compared to data transfer time. At the same time, they are concerned about how block size affects **MapReduce performance**. • Determine the **minimum block size** such that the disk seek time contributes to only **1% of the total data transfer time**. • Explain why HDFS typically uses **large block sizes (e.g., 128 MB or more)** in such distributed environments.	10	3	3

	• Analyze the impact on **MapReduce performance** if the block size is increased excessively.			
5	A climate research organization is analyzing historical weather data from the **National Climatic Data Center (NCDC)** to identify extreme temperature patterns. The dataset is very large (terabytes of data), stored in a distributed environment using **Apache Hadoop**. The researchers want to compute the **maximum temperature recorded for each year** to study climate change trends. How would you implement a MapReduce program to handle and process a large dataset efficiently using parallel processing?	10	3	1